

Mehnaz Rahman

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OBJECTIVE

Seeking a challenging technical management opportunity in wireless, machine learning, software, signal processing, IOT, algorithmic & architecture development, SOC and ASIC development. Technical hands-on experience on multiple areas of wireless communication, and solid R&D portfolio of innovations with multiple US patents and landmark publications in wireless communications.

Immigration Status US Citizen

SKILL SETS

Skills Wireless Communication (5G+, Wireless chip to chip communication, MIMO, Massive MIMO user scheduling using ML and GNN, K-best Decoder, prototyping, V2X communication), SOC and ASIC development, project technical leadership and management.

Leadership Acting as a coordinator and product owner between hardware and software group of almost 130 people at different geographical locations of the world, analyze and evaluate decision making, performance evaluation and also working closely with the business clients and their requirements.

Collaboration/Research Managing and handling projects with collaboration, workload distribution, code delivery etc. between Intel Santa Clara, CA, Hillsboro, OR and Mexico team.

EDUCATION

Ph. D. in Computer Engineering, Aug'12 to Aug'16

Texas A&M University, College Station, TX USA

GPA 3:80 out of 4:00

Thesis: An Iterative Soft Decision Based LR-Aided MIMO Decoder

B.S. in Electrical and Electronics Engineering (EEE), Jan'06 to Feb'11

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh

GPA 3:72 out of 4:00

COMPUTER SKILLS

Prog. Lang. C, MATLAB, python, SIMULINK, Verilog, Xilinx, Python, Perl

WORK EXPERIENCE

Jan'25- May'25 ML Certification from Cornell University and Stanford University,

Jan' 25- Present working as a TPM at stealth startup.

Aug'18 – Mar'23 Technical Program Manager at *Intel Research Lab*, Wireless Multi-Communication Research Lab (WMCR), Santa Clara, CA, USA.

MuMIMO with GNN: Working as a coordinator between algorithm and hardware team for GNN (Graph Neural Network) based multi-user MIMO scheduling. Responsibilities include modifying the algorithm to attain target performance, performance evaluation and required modification, training of GNN, test validation etc.

Chip to Chip THz Com.: Working as a coordinator and product owner between algorithm and hardware team for THz chip to chip communication. Responsibilities include modifying the algorithm to attain target performance, make it optimized and hardware friendly, validation, decision making, performance evaluation, and handing over the code to the hardware team for tape out.

Mmwave beamforming: Working as a product owner between teams of different geographical locations. Responsibilities include executing the project from generation of idea to final tape out, modifying the algorithm to make it optimized and hardware friendly, performance evaluation, validation, negotiation with the external client, Apple, and finally handover the project.

Aug '18 – Mar '23 Sr. Research Scientist at *Intel Research Lab*, Wireless and Communication Research Lab (WCR), Santa Clara, CA, USA.

MuMIMO with GNN: Pioneering project, includes algorithmic design, GNN training with reinforcement learning, test cases evaluation etc. Skill required: Matlab, python. Tensorflow.

Chip to Chip THz Com.: Pioneering project, includes algorithmic design, channel decoding and implementation of chip to chip Tera Hertz wireless communication. Skill required: Matlab.

Mmwave beamforming: Developed fixed point realization in order to implement and demonstrate mmwave multi user MIMO wireless communication on hardware. Responsibilities include modifying the algorithm to make it optimized and hardware friendly, decision making, performance evaluation and also working closely with the business clients and their requirements reporting directly to the director. Skill required: MATLAB, Verilog.

V2X Research: Developed a low-complexity and hardware efficient multi user MIMO detection algorithm considering V2X scenario and near far effects. The algorithm is scalable to any MIMO and user configuration. Skill required: MATLAB.

June '16 – Aug '18 Research Scientist at *Intel Research Lab*, Wireless and Communication Research Lab (WCR), Santa Clara, CA, USA.

Transformational Project includes implementing efficient communication Architecture & Design: algorithms using neural network. Built a customized neural network with I/O, and hidden layers; generated input and targeted output vectors from Markov Chain Monte Carlo (MCMC) code and trained the network accordingly to follow the targeted output pattern. Skill required: MATLAB.

5G Prototype: Generated input vectors, performed tests for Tx and FPGA interaction, and signal processing on Rx side for 1x1, 1x2, 2x1 and 2x2 MIMO. Skills required: C/C++, MATLAB, Verilog, and Perl.

Mmwave MIMO: Performed tests for 1 user 1 stream, 2 users 1 stream each and 1 user 2 streams MIMO configuration. Skills required: C/C++, MATLAB, and Perl.

May '15 – Aug '15, May '14 – Aug '14, May '13 – Aug '13 Graduate Technical Intern, *Intel Research Lab*, USA.

Performed stress testing on physical layer of LTE modem using C++ to verify L1C behavior and emulated PHY-PAL interaction, implemented 2G on Intel Architectures (MATLAB, C++ and Intel GPU) and achieved real time data rate as 277K symbols/sec using Intel GPU, and Submitted a patent on complex domain MIMO decoder and implemented it using C++ and MATLAB.

Aug '12 – May '16 Teaching Assistant *Texas A&M University, College Station, TX USA*.

Taught ECEN 248 during Fall and Spring 2013, Fall 2014, Spring and Fall 2015; ECEN 454/714 during Spring 2014 and 2016.

Awards/Certificate

- ML certificate from Cornell University and Stanford University
- WCR Team award 2019 at Intel Labs
- Intel Employee Recognition Award for 2017 and 2019 at Intel Labs
- Certificate of PMP Training Completion

PUBLICATIONS

Book *K-Best Decoders for 5G+ Wireless Communication*, M. Rahman, and Dr. G. Choi, in Springer International Publishing, Sept 2016.

Patent *High-isolation contactless waveguides (CWG) for enabling scalable 3D heterogeneous chiplet integration (submitted)*, Intel Lab.

Patent *Sub-THz Wireless Chip-to-Chip High Speed Data Transport Interconnect (submitted)*, Intel Lab.

Patent *Wireless chip to chip communication with selective frequency multiplexing with different modulation scheme (submitted)*, Intel Lab.

Patent *Method and devices for signal demodulation (submitted)*, Intel Lab.

Patent *Context aware synchronization method for decentralized V2V networks*, Sep 17, 2019, US10420051B2.

Patent *Joint equalization and non-linearity mitigation at the receiver with mixed signal processing (under review)*, Intel Labs.

Patent *Spatial multiplexing aided spatial modulation for MIMO systems (under review)*, Intel Lab.

Patent *Geolocation-based Pilot Sequence Assignment for De-centralized V2X Networks (under review)*, Intel Lab.

Patent *Multi-User Detection Friendly Pilot Design for Vehicle-to-Vehicle Communications (under review)*, Intel Lab.

Patent *Interworking of Vehicle-to-Anything (V2X) systems (under review)* with Intel Lab.

Patent *Priority-Based Collision-Avoidance Multi-User Access Scheme in Vehicle-to-Vehicle Communications*, Intel Lab.

For Other publications, please refer to

<https://www.linkedin.com/in/mehnaz-rahman-0b4b3a67/>

<https://scholar.google.com/citations?user=vvYwiokAAAAJ&hl=en>

REFERENCES

Anthony L. Chun (Tony), Platform Architect, tonylchun@gmail.com